

Supplementary information of: "Review of Experimental Data and Evaluation of Equations of State for Modeling Formation of Solid CO₂ in CCS and Natural Gas Applications"

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In the Supplementary Information (SI) we present information about the binary interaction parameters used in Sec. 1, a comparison of the performance of tc-PR EoS before and after modifying the α -function in Sec. 2, and additional comparisons and figures with solid-fluid equilibrium (SFE) predictions in Sec. 3.

1 Binary interaction parameters of the different equations of state

Table SI.1 lists the binary interaction parameters used in this work, k_{ij} . They have all been regressed based on the same experimental vapor-liquid equilibrium data, and the resulting vapor-liquid equilibrium envelopes are shown in Figures SI.1, SI.2, and SI.3.

Table SI.1: List of k_{ij} parameters for a variety of component pairs.

Pair	PR/tc-PR	SRK	PC-SAFT	PCP-SAFT	SAFT-VR Mie
Ar/CO ₂	0.132091	0.131588	0.083453	0.013313	0.008156
CH ₄ /C ₂ H ₆	0.005330	-0.000195	-0.000531	-0.000531	-0.014625
CH ₄ /C ₃ H ₈	0.009800	0.002700	0.005438	0.005438	-0.007500
CH ₄ /CO ₂	0.123000	0.121000	0.058180	-0.120000	0.024000
CH ₄ /H ₂	-0.191813	-0.287750	0.014250	0.014250	-0.445125
CH ₄ /N ₂	0.038145	0.037294	0.035406	0.026297	0.019766
CH ₄ /O ₂	0.075000	0.083000	0.065813	0.065813	0.022063
CH ₄ /C ₄ H ₁₀	0.032564	0.025550	0.016875	0.016875	0.010438
C ₂ H ₆ /C ₃ H ₈	-0.008910	-0.009460	-0.004875	-0.004875	-0.006375
C ₂ H ₆ /CO ₂	0.120293	0.124219	0.084787	0.010120	0.053063
C ₃ H ₈ /CO ₂	0.114375	0.119121	0.093023	0.021534	0.058625
CO ₂ /H ₂	0.315738	0.334181	0.154250	0.076875	-0.466875
CO ₂ /H ₂ S	0.097453	0.099844	0.053241	-0.008641	-0.142375
CO ₂ /N ₂	0.012066	-0.002072	0.022156	-0.021625	-0.077656
CO ₂ /O ₂	0.136345	0.136053	0.080094	0.008313	-0.018250
CO ₂ /SO ₂	0.064181	0.008709	-0.178875	-0.178875	-0.032688
CO ₂ /C ₄ H ₁₀	0.104121	0.108633	0.098739	0.026228	0.059813
N ₂ /O ₂	-0.020825	-0.031350	-0.004875	-0.010875	-0.186813

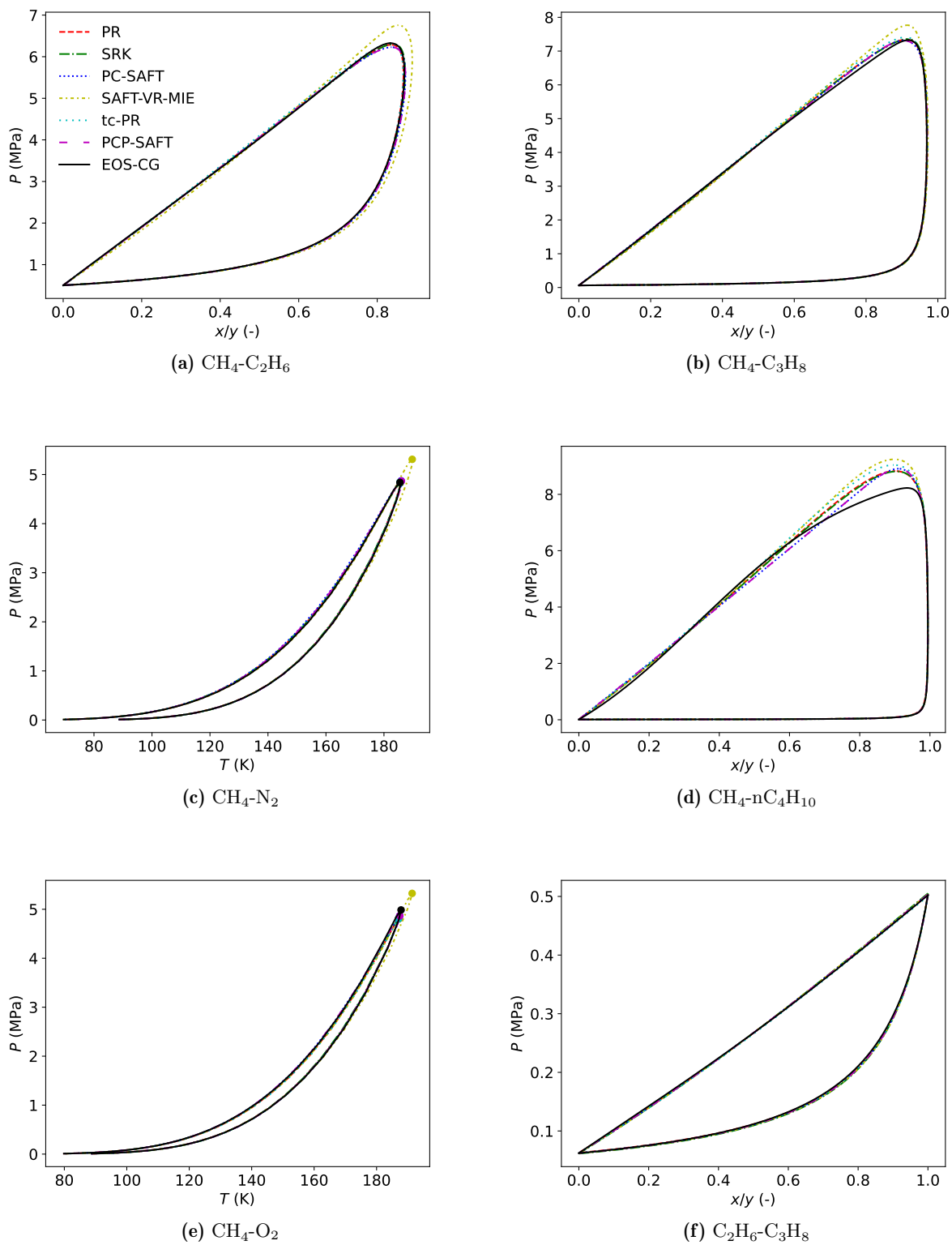


Figure SI.1: Illustration of phase envelopes of different binary mixtures after regressing interaction parameters. The curves correspond to the legends in Subfigure SI.1a.

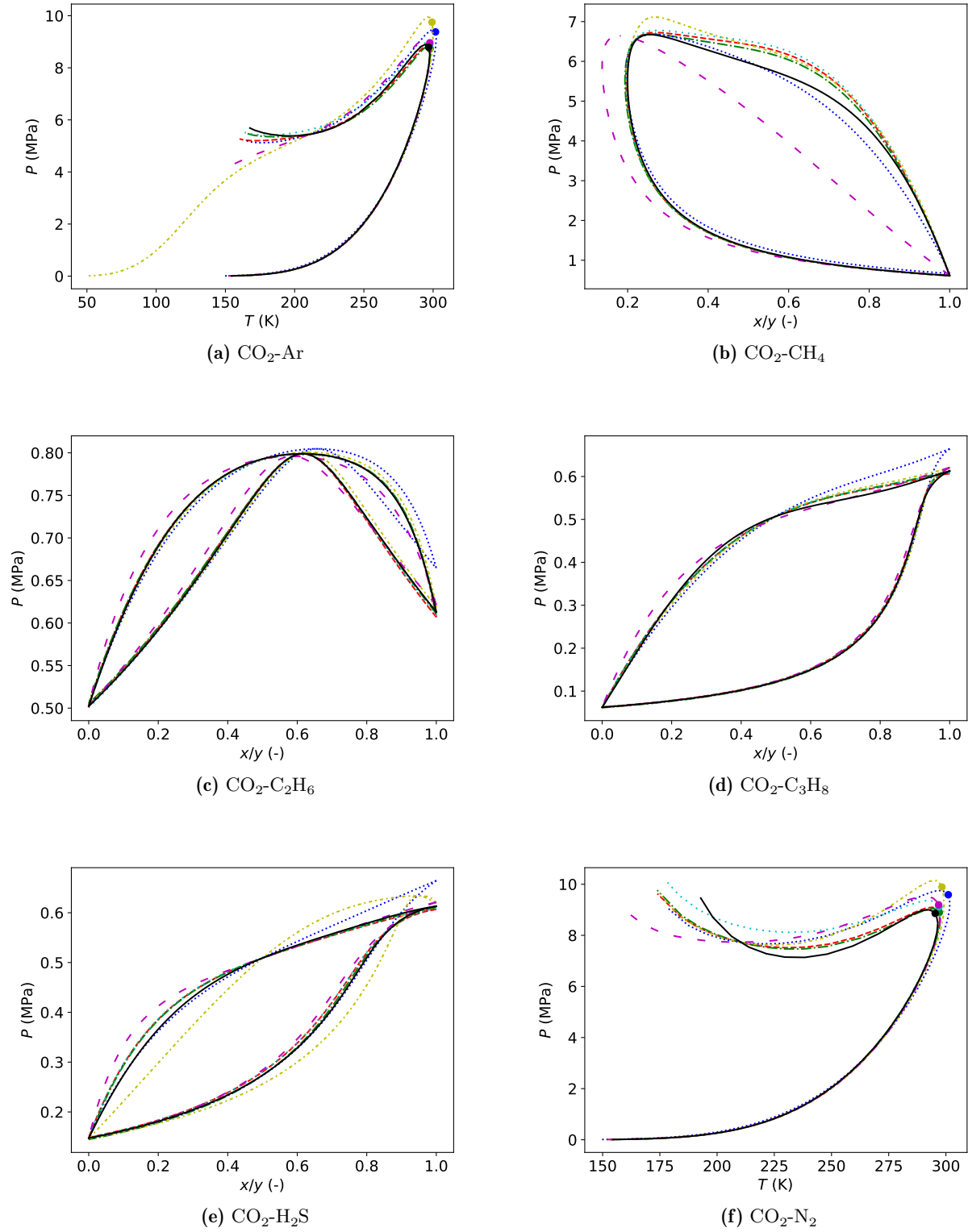


Figure SI.2: Illustration of phase envelopes of different binary mixtures after regressing interaction parameters. The curves correspond to the legend in Subfigure SI.1a.

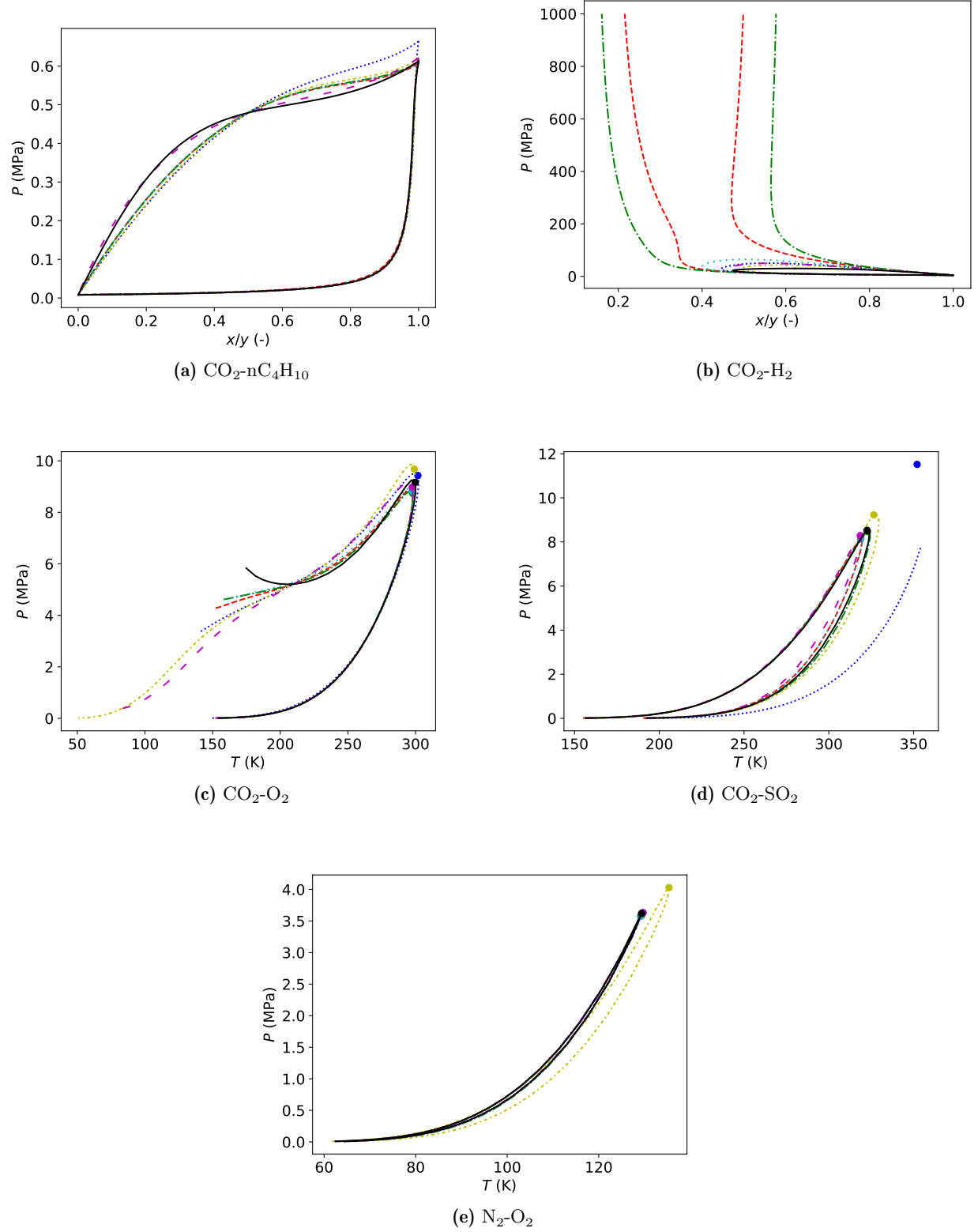


Figure SI.3: Illustration of phase envelopes of different binary mixtures after regressing interaction parameters. The curves correspond to the legend in Subfigure SI.1a.

2 tc-PR comparisons with modified parameters

The α -function of tc-PR was modified by including experimental data below the triple point, as explained in the article. The Mean Absolute Percentage Deviations (MAPD) for the properties along the vapor-liquid coexistence curve (subscript "sat") for tc-PR before and after the modification are shown in Tab. SI.2. The table suggests that by including one more parameter in the temperature-dependent expression for the α -function, it might be possible to further improve the performance.

Table SI.2: MAPDs for thermophysical properties compared with properties from EoS-CG along the vapor-liquid equilibrium curve. The evaluated properties include the saturation pressure (p_{sat}), density (ρ_{sat}), vaporization enthalpy (Δh_{vap}) and heat capacity ($c_{p,sat}$).

	p_{sat}	ρ_{sat}	Δh_{vap}	$c_{p,sat}$
tc-PR	0.3063	2.0404	1.5735	11.3863
tc-PR*	0.3724	2.4659	2.0903	12.5166

3 Additional SVE and SLE predictions

The relative deviations in melting temperature predictions of pure CO₂ is shown in Fig. SI.4, up to the highest pressures seen in the experiments for mixtures. Solid Vapor Equilibrium (SVE) of the CO₂/CH₄ mixture is shown in Fig SI.5. Additional Solid Liquid Equilibrium (SLE) predictions for CO₂/N₂ are shown in Figure SI.6. Residual figures for all the EoS for the CO₂/CH₄ SVE system are shown in Fig. SI.7 and Fig. SI.8. Figure SI.9 shows the residual temperature for SLE for different binaries for the EoS not shown in the main article. For both PC-SAFT and SAFT-VR Mie, some of the residuals in Figure SI.9 were too large to be included, e.g. for CO₂/H₂S.

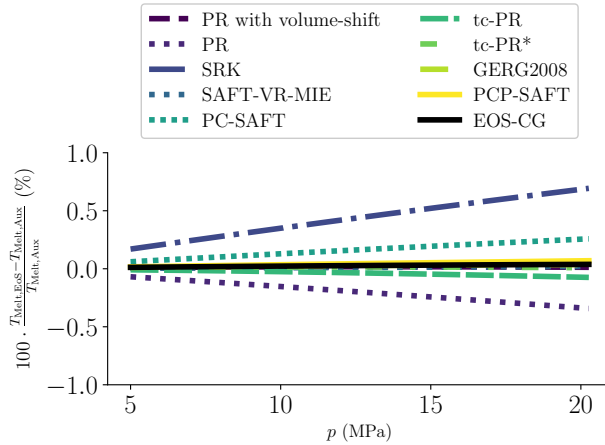


Figure SI.4: Relative deviations in the melting temperature of pure CO₂, plotted as a function of pressures up to the maximum pressure seen in the mixture experiments (20.2650 MPa).

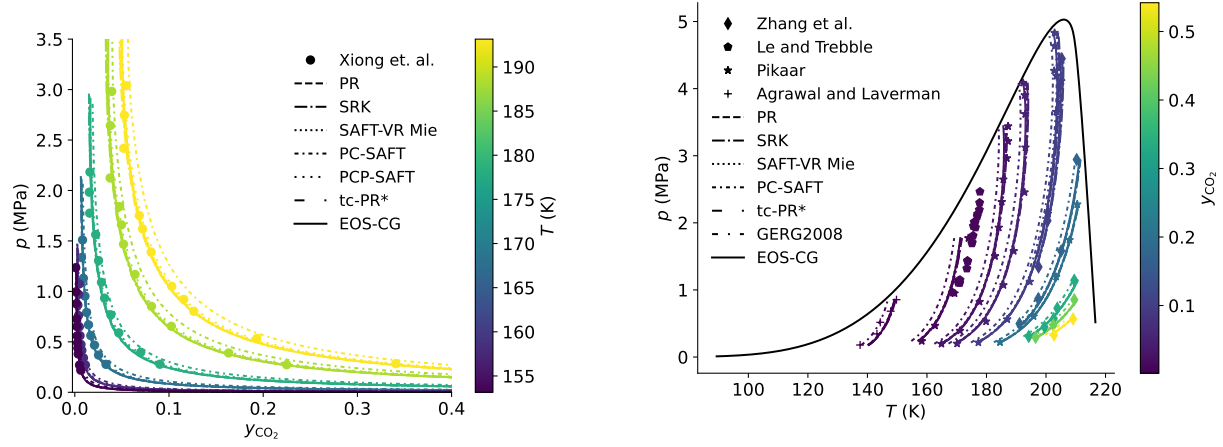


Figure SI.5: SVE equilibrium of the CO₂/CH₄ system for a set of isotherms (left), and a set of constant fraction lines (right), compared with measurements [1–5].

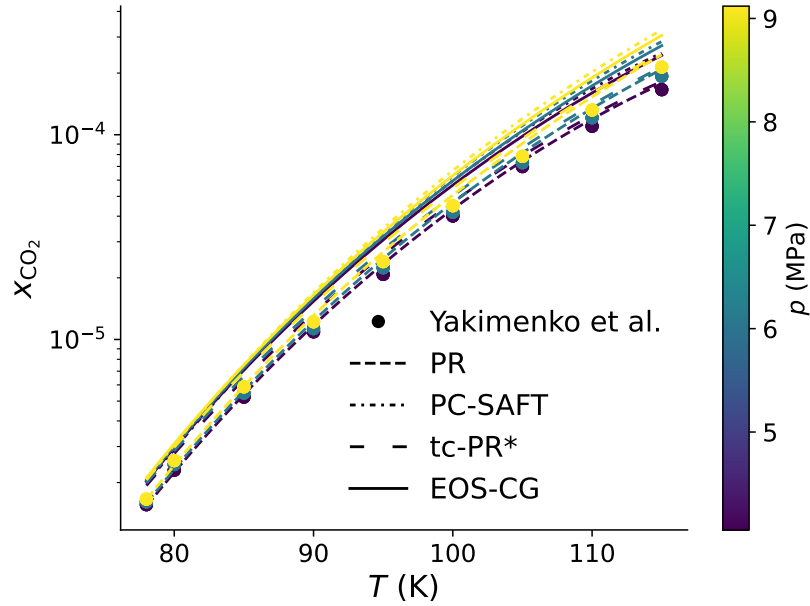


Figure SI.6: SLE of carbon dioxide and nitrogen compared with measurements [6].

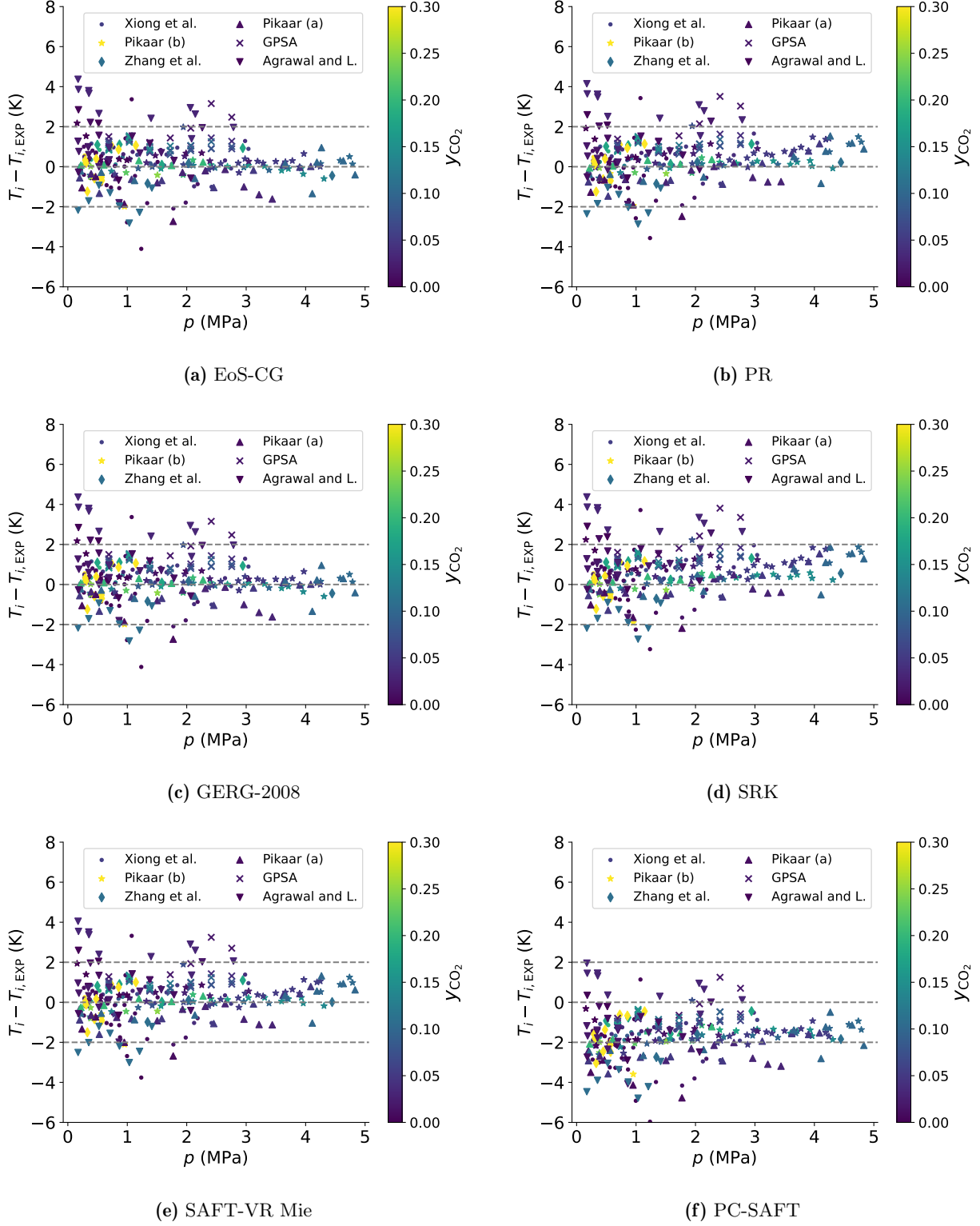


Figure SI.7: Residual predicted temperature between measurements [1, 2, 4, 5, 7] for SVE equilibrium of a CO_2/CH_4 system. "a" and "b" refers to the measuring devices Pikaar [4] utilized in their work. The data of Le and Trebble [3] are not included as these are noted to be inconsistent in Ref. [8].

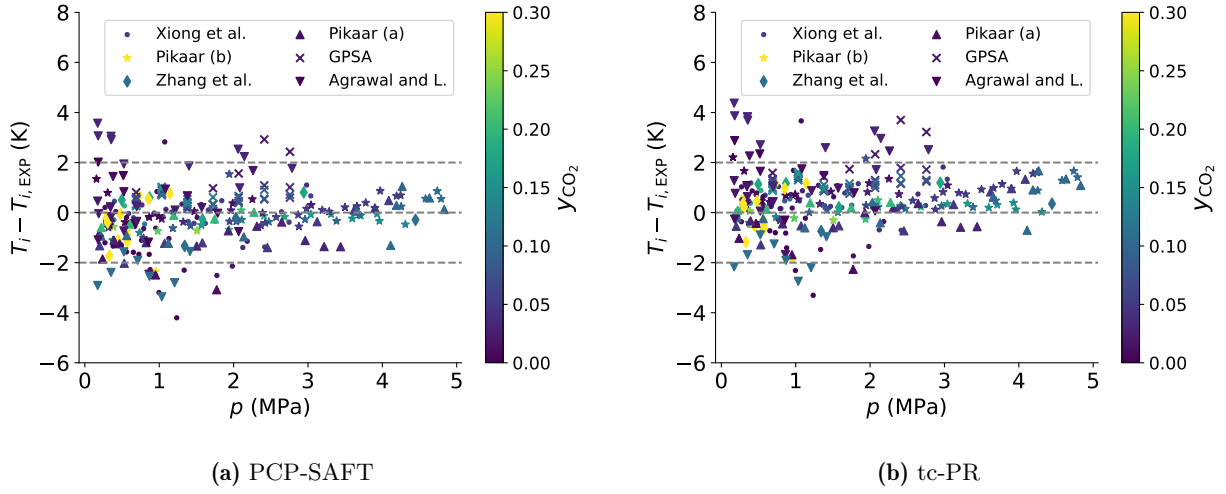
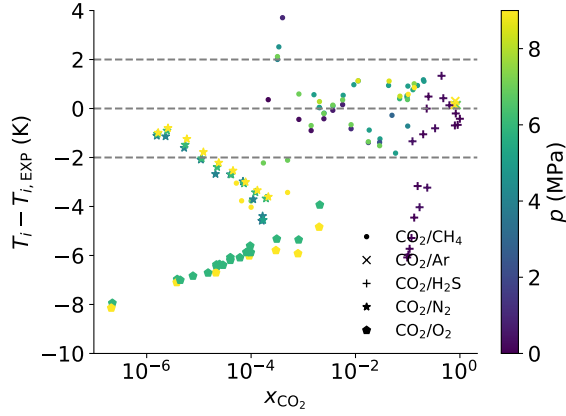
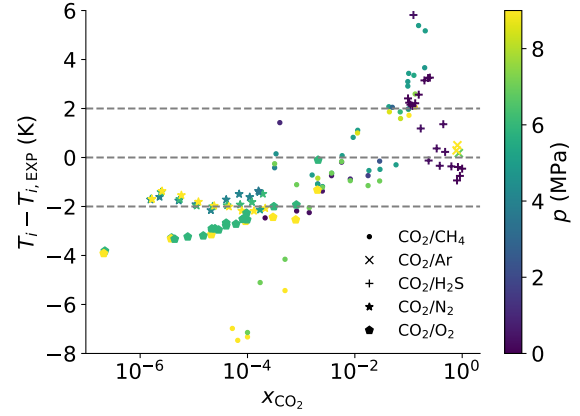


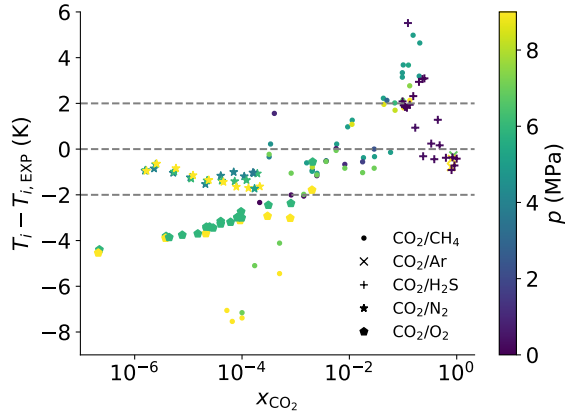
Figure SI.8: Residual predicted temperature compared with measurements [1,2,4,5,7] for SVE equilibrium of a CO_2/CH_4 system. "a" and "b" refers to the measuring devices Pikaar [4] utilized in their work. The data of Le and Trebble [3] are not included as these are noted to be inconsistent in Ref. [8].



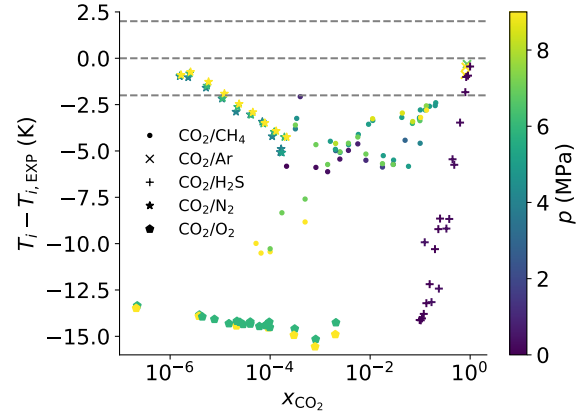
(a) GERG-2008



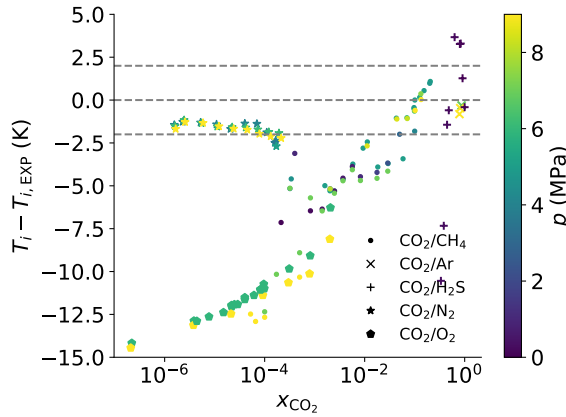
(b) SRK



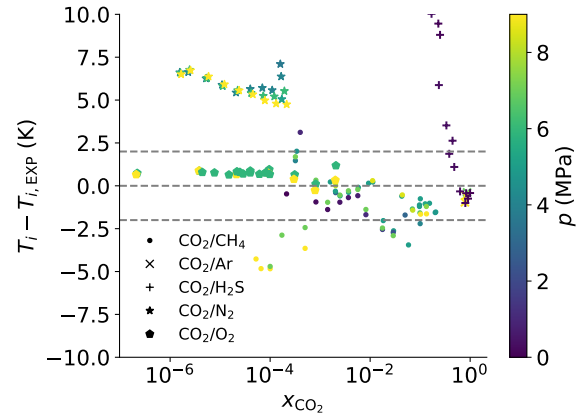
(c) tc-PR



(d) PC-SAFT



(e) SAFT-VR Mie



(f) PCP-SAFT

Figure SI.9: Residual predicted temperature compared with measurements of SLE equilibrium [4, 6, 9–16] for different EoS'.

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